

```

amir@Legion-5-Pro-16ITH6:~$ sudo mn --topo single,5 --controller=remote,ip=127.0.0.1,port=6633
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3 h4 h5
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1) (h4, s1) (h5, s1)
*** Configuring hosts
h1 h2 h3 h4 h5
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> 

```

openflow_v4

No.	Time	Source	Destination	Protocol	Length	Info
11	0.245549001	127.0.0.1	127.0.0.1	OpenFL...	82	Type: OFPT_HELLO
15	0.259587222	127.0.0.1	127.0.0.1	OpenFL...	146	Type: OFPT_PORT_STATUS
✓ 17	0.262996595	127.0.0.1	127.0.0.1	OpenFL...	74	Type: OFPT_FEATURES_REQUEST
18	0.263159047	127.0.0.1	127.0.0.1	OpenFL...	98	Type: OFPT_FEATURES_REPLY

```

▶ Frame 17: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface lo, id 0
▶ Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
▶ Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
▶ Transmission Control Protocol, Src Port: 6633, Dst Port: 54886, Seq: 17, Ack: 89, Len: 8
▼ OpenFlow 1.3
  Version: 1.3 (0x04)
  Type: OFPT_FEATURES_REQUEST (5)
  Length: 8
  Transaction ID: 2

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> Frame 18: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface lo, id 0
> Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
> Transmission Control Protocol, Src Port: 54886, Dst Port: 6633, Seq: 89, Ack: 25, Len: 32
- OpenFlow 1.3
  Version: 1.3 (0x04)
  Type: OFPT_FEATURES_REPLY (6)
  Length: 32
  Transaction ID: 2
  datapath_id: 0x0000000000000001
  n_buffers: 0
  n_tables: 254
  auxiliary_id: 0
  Pad: 0
  capabilities: 0x0000004f
  Reserved: 0x00000000

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> Frame 69: 198 bytes on wire (1584 bits), 198 bytes captured (1584 bits) on interface lo, id 0
> Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
> Transmission Control Protocol, Src Port: 54886, Dst Port: 6633, Seq: 9121, Ack: 1086, Len: 132
- OpenFlow 1.3
  Version: 1.3 (0x04)
  Type: OFPT_PACKET_IN (10)
  Length: 132
  Transaction ID: 0
  Buffer ID: OFP_NO_BUFFER (4294967295)
  Total length: 90
  Reason: OFPR_NO_MATCH (0)
  Table ID: 0
  Cookie: 0x000000000000000a
  Match
  Pad: 0000
  Data

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66	1.293784137	127.0.0.1	127.0.0.1	OpenFl...	74 Type: OFPT_BARRIER_REQUEST
67	1.294043195	127.0.0.1	127.0.0.1	OpenFl...	74 Type: OFPT_BARRIER_REPLY
69	1.398214944	127.0.0.1	127.0.0.1	OpenFl...	198 Type: OFPT_PACKET_IN
70	1.398260049	127.0.0.1	127.0.0.1	OpenFl...	178 Type: OFPT_PACKET_IN
72	1.399197131	127.0.0.1	127.0.0.1	OpenFl...	198 Type: OFPT_PACKET_IN
73	1.399228313	127.0.0.1	127.0.0.1	OpenFl...	178 Type: OFPT_PACKET_IN
75	1.462331050	127.0.0.1	127.0.0.1	OpenFl...	198 Type: OFPT_PACKET_IN
76	1.462376961	127.0.0.1	127.0.0.1	OpenFl...	178 Type: OFPT_PACKET_IN
78	1.719169554	127.0.0.1	127.0.0.1	OpenFl...	198 Type: OFPT_PACKET_IN
80	1.719224970	127.0.0.1	127.0.0.1	OpenFl...	178 Type: OFPT_PACKET_IN
82	1.822213144	127.0.0.1	127.0.0.1	OpenFl...	198 Type: OFPT_PACKET_IN
84	2.166186514	127.0.0.1	127.0.0.1	OpenFl...	198 Type: OFPT_PACKET_IN
86	2.166236715	127.0.0.1	127.0.0.1	OpenFl...	198 Type: OFPT_PACKET_IN
88	2.166272787	127.0.0.1	127.0.0.1	OpenFl...	178 Type: OFPT_PACKET_IN
90	2.166447439	127.0.0.1	127.0.0.1	OpenFl...	198 Type: OFPT_PACKET_IN
92	2.486402763	127.0.0.1	127.0.0.1	OpenFl...	198 Type: OFPT_PACKET_IN
94	2.806431787	127.0.0.1	127.0.0.1	OpenFl...	198 Type: OFPT_PACKET_IN

```

mininet> h1 ping h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=1.12 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.034 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.064 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.087 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.038 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.087 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.027 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.027 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.035 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.074 ms
^C

```

```

amir@Legion-5-Pro-16ITH6:~$ sudo mn --topo single,5 --controller remote -x
*** Creating network
*** Adding controller
Connecting to remote controller at 127.0.0.1:6653
*** Adding hosts:
h1 h2 h3 h4 h5
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1) (h4, s1) (h5, s1)
*** Configuring hosts
h1 h2 h3 h4 h5
*** Running terms on :1
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> 

```

```

amir@Legion-5-Pro-16ITH6:~$ sudo mn --topo single,5 --mac --switch ovsk,protocols=OpenFlow13 --controller=remote,ip=127.0.0.1,port=6633
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3 h4 h5
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1) (h4, s1) (h5, s1)
*** Configuring hosts
h1 h2 h3 h4 h5
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> 

```

```
amir@Legion-5-Pro-16ITH6:~$ sudo mn --topo single,5 --controller remote -x
*** Creating network
*** Adding controller
Connecting to remote controller at 127.0.0.1:6653
*** Adding hosts:
h1 h2 h3 h4 h5
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1) (h4, s1) (h5, s1)
*** Configuring hosts
h1 h2 h3 h4 h5
*** Running terms on :1
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> 
```

```
amir@Legion-5-Pro-16ITH6:~$ sudo mn --topo tree,5 --mac --arp --controller=remote,ip=127.0.0.1,port=6633
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h21 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32
*** Adding switches:
s1 s2 s3 s4 s5 s6 s7 s8 s9 s10 s11 s12 s13 s14 s15 s16 s17 s18 s19 s20 s21 s22 s23 s24 s25 s26 s27 s28 s29 s30 s31
*** Adding links:
(s1, s2) (s1, s17) (s2, s3) (s2, s10) (s3, s4) (s3, s7) (s4, s5) (s4, s6) (s5, h1) (s5, h2) (s6, h3) (s6, h4) (s7, s8) (s7, s9) (s8, h5) (s8, h6) (s9, h7) (s9, h8) (s10, s11) (s10, s14) (s11, s12)
(s11, s13) (s12, h9) (s12, h10) (s13, h11) (s13, h12) (s14, s15) (s14, s16) (s15, h13) (s15, h14) (s16, h15) (s16, h16) (s17, s18) (s17, s25) (s18, s19) (s18, s22) (s19, s20) (s19, s21) (s20, h17)
(s20, h18) (s21, h19) (s21, h20) (s22, s23) (s22, s24) (s23, h21) (s23, h22) (s24, h23) (s24, h24) (s25, s26) (s25, s29) (s26, s27) (s26, s28) (s27, h25) (s27, h26) (s28, h27) (s28, h28) (s29, s30)
(s29, s31) (s30, h29) (s30, h30) (s31, h31) (s31, h32)
*** Configuring hosts
h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14 h15 h16 h17 h18 h19 h20 h21 h22 h23 h24 h25 h26 h27 h28 h29 h30 h31 h32
*** Starting controller
c0
*** Starting 31 switches
s1 s2 s3 s4 s5 s6 s7 s8 s9 s10 s11 s12 s13 s14 s15 s16 s17 s18 s19 s20 s21 s22 s23 s24 s25 s26 s27 s28 s29 s30 s31 ...
*** Starting CLI:
mininet> 
```



```

done
amir@Legion-5-Pro-16ITH6:~$ sudo mn --topo linear --controller=remote,ip=127.0.0.1,port=6633
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1 s2
*** Adding links:
(h1, s1) (h2, s2) (s2, s1)
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 2 switches
s1 s2 ...
*** Starting CLI:
mininet> 

```

```

*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14
*** Adding switches:
s1 s2 s3 s4 s5 s6 s7 s8
*** Adding links:
(h1, s1) (h2, s2) (h3, s2) (h4, s3) (h5, s4) (h6, s4) (h7, s4) (h8, s5) (h9, s6) (h10, s6) (h11, s7) (h12, s7) (h13, s8) (h14, s8)
*** Configuring hosts
h1 h2 h3 h4 h5 h6 h7 h8 h9 h10 h11 h12 h13 h14
*** Starting controller
c0
*** Starting 8 switches
s1 s2 s3 s4 s5 s6 s7 s8

*** Dumping network
*** Switches:
- s1
- s2
- s3
- s4
- s5
- s6
- s7
- s8

*** Hosts:
- h1
- h2
- h3
- h4
- h5
- h6
- h7
- h8
- h9
- h10
- h11
- h12
- h13
- h14

*** Links:
- s1 ↔ s2
- s2 ↔ s3
- s3 ↔ s4
- s5 ↔ s6
- s6 ↔ s7
- s7 ↔ s8
- s2 ↔ s8

*** Nodes:
Nodes in the network:
- h1
- h2
- h3

```

```
- s7  
- s8
```

*** Hosts:

```
- h1  
- h2  
- h3  
- h4  
- h5  
- h6  
- h7  
- h8  
- h9  
- h10  
- h11  
- h12  
- h13  
- h14
```

*** Links:

```
- s1 ↔ s2  
- s2 ↔ s3  
- s3 ↔ s4  
- s5 ↔ s6  
- s6 ↔ s7  
- s7 ↔ s8  
- s2 ↔ s8
```

*** Nodes:

Nodes in the network:

```
- h1  
- h2  
- h3  
- h4  
- h5  
- h6  
- h7  
- h8  
- h9  
- h10  
- h11  
- h12  
- h13  
- h14  
- s1  
- s2  
- s3  
- s4  
- s5  
- s6  
- s7  
- s8
```

*** Testing network connectivity (pingall):

Pingall result: 0 (All hosts are reachable)

amir@Legion-5-Pro-16ITH6:~/Desktop\$